

Possible Extensions of GLM's

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1 Generalized Linear Mixed Models

The Mathematical expression of GLMM is

$$h(E(y_i)) = \mathbf{X}_i\boldsymbol{\beta} + W_i\mathbf{r}_i, \quad i = 1, 2, \dots, n \quad (1)$$

$$\mathbf{r}_i \sim N(0, D^*), \quad (2)$$

where $y_i = (y_{i1}, y_{i2}, \dots, y_{ini})$ are the repeated measurements within individual or cluster i , $h(\cdot)$ is a known monotone link function, $\mathbf{X}_i$ and W_i are known design matrices, $\boldsymbol{\beta}$ contains fixed effects, $\mathbf{r}_i$ contains random effects, and D^* is a covariance matrix. It is typically assumed that the responses y_{ij} are conditionally independent and follow distributions in the exponential family, given the random effects.

Statistical inference for a GLMM is based on the likelihood method. The marginal distribution for y_i is

$$f(\mathbf{y}_i|\boldsymbol{\beta}, D^*) = \int \prod_{j=1}^{n_i} [f(y_{ij}|\mathbf{x}_{ij}, w_{ij}, \boldsymbol{\beta}, \phi, \mathbf{r}_i) f(\mathbf{r}_i|D^*)] d\mathbf{r}_i \quad (3)$$

which is not in closed-form expression as the model is nonlinear in the random effects $\mathbf{r}_i$. The likelihood for all observed data is given by

$$L(\boldsymbol{\beta}, D^*|\mathbf{y}_i) = \prod_{i=1}^n \left\{ \int \prod_{j=1}^{n_i} [f(y_{ij}|\mathbf{x}_{ij}, w_{ij}, \boldsymbol{\beta}, \phi, \mathbf{r}_i) f(\mathbf{r}_i|D^*)] d\mathbf{r}_i \right\} \quad (4)$$

Likelihood inference for a GLMM can be difficult due multi-dimensional integral with respect to the random effects. The estimation of equation (7) based on the following methods **Monte Carlo EM** algorithms, **Numerical integration methods** such as the Gaussian Hermite quadrature method, **Approximate methods** based on Laplace approximations or Taylor approximations.

Note 1:

- A **Monte Carlo** EM algorithm can be used by treating the random effects $\mathbf{r}_i$ as missing data, but it sometimes can have **convergence problems**.
- **Numerical integration** methods are often used when the dimension of the integral is low. ie **maximum 2**.
- **Approximate methods** are computationally very **efficient**, but the accuracy of the approximation sometimes may not be satisfactory.

Note 2: Logistic regression model with random effects: Consider a longitudinal binary response y_{ij} taking only two possible values (say, 0 or 1), $i = 1, \dots, n$; $j = 1, \dots, n_i$. A simple logistic regression model with random intercept as below

$$\begin{aligned} \log\left(\frac{\mu_{ij}}{1 - \mu_{ij}}\right) &= \beta_{0i} + \beta_1 t_{ij} = (\beta_0 + \mathbf{r}_i) + \beta_1 t_{ij}, \\ \mathbf{r}_i &\sim N(0, d^{*2}), \end{aligned}$$

where $\mu_{ij} = E(y_{ij}) = P(y_{ij} = 1)$ and $\beta_{0i} = \beta_0 + \mathbf{r}_i$.

A **general form** of GLMM for binary longitudinal or clustered responses may be written as

$$\begin{aligned} \log\left(\frac{\mu_{ij}}{1 - \mu_{ij}}\right) &= \mathbf{X}_{ij}^\top \boldsymbol{\beta} + \mathbf{W}_{ij}^\top \mathbf{r}_i, \\ \mathbf{r}_i &\sim N(0, D^*) \end{aligned}$$

Note 3: Poisson regression model with random effects. For longitudinal or clustered count responses y_{ij} , we consider the Poisson regression models with random effects as below

$$\begin{aligned} \log(\mu_{ij}) &= \mathbf{X}_{ij}^\top \boldsymbol{\beta} + W_{ij}^\top \mathbf{r}_i, \\ \mathbf{r}_i &\sim N(0, D^*) \end{aligned}$$

where $\mu_{ij} = E(y_{ij})$.

Example 2. bacteria data set is related to the presence of Bacteria after Drug Treatments. More information about the bacteria data set can be found in the MASS R package. solution:

1. Here is the part of data.

```
> data("bacteria")
> head(bacteria)
  y ap hilo week  ID      trt
1 y p  hi    0 X01 placebo
2 y p  hi    2 X01 placebo
3 y p  hi    4 X01 placebo
4 y p  hi   11 X01 placebo
5 y a  hi    0 X02  drug+
6 y a  hi    2 X02  drug+
.....
```

2. Firstly, We fit GLM that ignores **longitudinal correlation** and treat all data as independent identical distributed (i.i.d) and the below outcome

```
> glm1=glm(y ~ trt + I(week > 2), family = binomial, data = bacteria)
> summary(glm1)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	2.8332	0.4506	6.287	3.24e-10	***
trtdrug	-1.1187	0.4288	-2.609	0.00909	**
trtdrug+	-0.6372	0.4487	-1.420	0.15555	
I(week > 2)TRUE	-1.2949	0.4104	-3.155	0.00160	**

3. Now, we fit **GLMM** which uses random effects to incorporate **longitudinal correlation over time**. We use R function **glmmPQL** that is based on an **approximate method** for GLMMs. This is case 1 for **random effect** on the **intercept**. We found that the **estimates** of GLMM differ from GLM fit, when **correlation** is incorporated and see below:

```
> glmm1=glmmPQL(y ~ trt + I(week > 2), random = ~ 1 | ID,
                 family = binomial, data = bacteria)
> summary(glmm1)
Fixed effects: y ~ trt + I(week > 2)
              Value Std.Error DF   t-value p-value
(Intercept)   3.412014 0.5185033 169   6.580506  0.0000
trtdrug       -1.247355 0.6440635  47  -1.936696  0.0588
trtdrug+      -0.754327 0.6453978  47  -1.168779  0.2484
I(week > 2)TRUE -1.607257 0.3583379 169  -4.485311  0.0000
```

4. We consider case 2 when random effect on intercept and trt.

```
glmm2=glmmPQL(y ~ trt + I(week > 2), random = ~ 1+trt | ID,  
              family = binomial, data = bacteria)  
> summary(glmm2)  
Fixed effects: y ~ trt + I(week > 2)  
              Value Std.Error DF   t-value p-value  
(Intercept)   3.553532 0.5818242 169   6.107570  0.0000  
trtdrug        -1.340990 0.7204088  47  -1.861429  0.0689  
trtdrug+       -1.020495 0.6116403  47  -1.668456  0.1019  
I(week > 2)TRUE -1.629387 0.3640741 169  -4.475427  0.0000
```